Voltage Stability Assessment Methods in Large-Scale Power Systems

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Abstract

Voltage stability has become a major operational challenge for modern large-scale power systems, especially with the growing penetration of inverter and renewable based energy resources. Insufficient reactive-power support and stressed network conditions can lead to voltage collapse and blackouts. This paper presents a comprehensive review of voltage-stability assessment methods, ranging from classical deterministic techniques such as P–V and Q–V curve analysis, continuation power flow (CPF), and modal analysis to advanced probabilistic, synchrophasor-based, and machine-learning approaches.Their theoretical foundation, computational features, and industry uptake are highlighted. The review ends by outlining new approaches to real-time, data-driven, hybrid voltage-stability assessment that are appropriate for the changing power grid.

Keywords

Voltage Stability, P–V Curve, Q–V Curve, Continuation Power Flow, Modal Analysis, Voltage Collapse, Inverter-Based Resources, Machine Learning, Probabilistic Assessment.

1. Introduction

Voltage stability is a fundamental aspect of power-system reliability, ensuring that all bus voltages remain within acceptable levels during steady-state operation and following disturbances [1]. From a physical standpoint, voltage instability occurs when the system cannot maintain equilibrium between reactive-power generation and consumption, so that incremental increases in load or disturbances cause progressive voltage decline, which may ultimately lead to voltage collapse if no corrective actions are taken [2].

In modern large-scale networks, the importance of voltage stability has increased with stronger interconnections, heavily loaded transmission corridors, and higher penetration of renewable generation. Inverter-based resources (IBRs) such as wind and solar photovoltaic plants typically contribute less inherent reactive-power support and exhibit different dynamic characteristics than conventional synchronous machines, which can erode voltage margins if they are not coordinated with shunt compensation, on-load tap changers, and excitation systems [4].

In this context, engineers and operators are confronted with a wide toolbox of voltage-stability techniques, each built on different assumptions, data requirements, and time scales. Deterministic power-flow–based tools, probabilistic methods, synchrophasor-driven indicators, and data-driven algorithms often coexist in planning software and control-room applications, yet their relative strengths and blind spots are not always clear from a practical perspective. This paper reviews how these methods relate conceptually, indicates which planning and operating questions each can address, and outlines emerging hybrid frameworks for renewable-rich, increasingly stressed power systems.

Case Study:

On August 14, 2003, the Northeast blackout provided a prototypical example of voltage instability escalating into a wide-area cascade. In the FirstEnergy system in Ohio, high loading, several generators and capacitor banks out of service, and weak transmission corridors left limited reactive-power reserves and poor voltage control. As a result successive line trips and growing reactive losses caused progressive voltage depression on key 345 kV buses. Meanwhile, operators did not shed load or reconfigure the system in time, allowing low voltages and thermal overloads to propagate, trip

additional lines and generators, and island parts of the Eastern Interconnection, ultimately interrupting supply to tens of millions of customers. The U.S.–Canada Task Force report highlighted technical remedies including stronger NERC criteria and enforcement for reactive-power and voltage-control planning, enhanced real-time monitoring and improved protection to prevent similar cascades [3].

2. Fundamentals of Voltage Stability

Voltage instability arises due to the nonlinear relationship between power and voltage magnitude. At high system loading, reactive demand increases and voltages fall until the network can no longer maintain power balance. The Jacobian matrix of the load flow becomes singular at this condition:

$$det\ (Jqv)\ =\ 0$$

At this critical point, even small load increments cause disproportionate voltage drops, resulting in collapse.

Voltage stability is typically classified as:

- Static (steady-state): analysis using algebraic power-flow equations; and
- Dynamic: analysis including load, excitation, and control system dynamics. [5]

3. Literature Review

3.1 Power System Stability and Control by Kundur (2004).

Power System Stability and Control by Kundur (2004) explains voltage stability assessment in P–V Curve which is also called the *nose curve*, shows the variation of bus voltage with increasing active power. It is used to find the voltage-collapse point and the stability margin—the distance from the current operating point to the maximum load-ability.

A simple two-bus power-voltage relationship is:

$$P = \frac{V}{X}(V_s cos\ \delta - V)$$

Stable region condition (upper branch):

$$\frac{dV}{dP} < 0$$

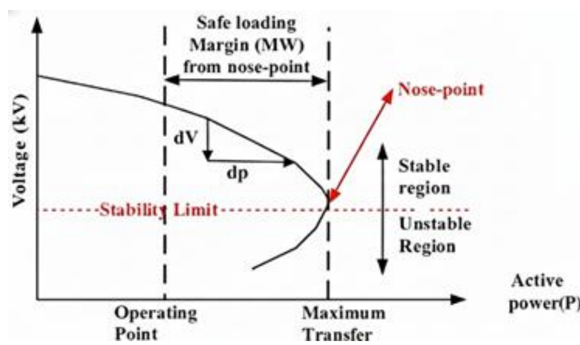
Unstable region (lower branch):

$$\frac{dV}{dP} > 0$$

At the nose point:

$$\frac{dV}{dP} = 0$$

Fig. 1. *Typical P–V curve showing stable and unstable regions.*



The upper side branch of the curve represents stable operation, while the lower branch is unstable. The "nose point" it's also called turning point,which indicates the maximum loading capability of the system [1].

3.2 Power System Voltage Stability by Taylor (1994).

Power System Voltage Stability by Taylor (1994) explains voltage stability assessment using the Q–V curve which plots the relationship between reactive power (Q) injection and voltage (V) at a given bus. The minimum point of the curve identifies the critical voltage and defines the reactive-power margin. [2]

The fundamental relationship at a load bus:

$$Q = \frac{V}{X}(V - V_s cos\ \delta)$$

Stable region:

$$\frac{dQ}{dV} > 0$$

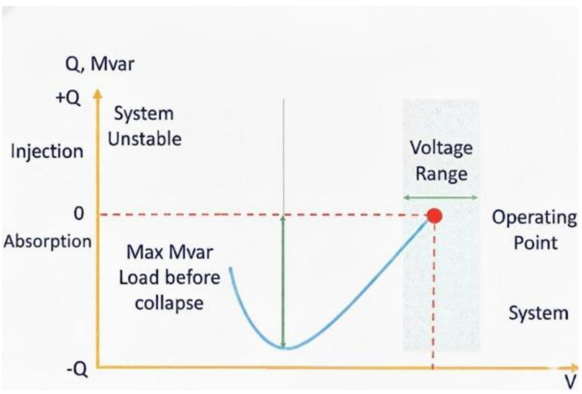
Unstable region:

$$\frac{dQ}{dV} < 0$$

Critical (minimum point):

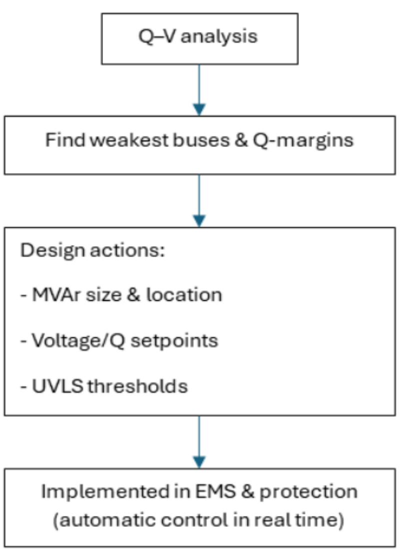
$$\frac{dQ}{dV} = 0$$

Fig. 2. *Illustrative Q–V curve.*



Q–V analysis is widely used in practice to assess voltage stability and determine how much reactive support the system requires under stressed conditions. It helps operators identify weak buses and plan corrective actions before voltage collapse occurs.

Fig. 3. *Application of Q–V curve.*



The workflow is summarized in the flowchart shown here. The process begins with performing an offline Q–V analysis to evaluate system voltage stability. The results are then used to identify the weakest buses—those with the lowest Q-margins. Once these critical locations are known, planners determine the required reactive support, such as the MVAr rating and placement of capacitors, SVCs, or STATCOMs, along with appropriate generator voltage and Q setpoints and UVLS thresholds. Finally, these settings and corrective actions are implemented in the EMS and protection systems to ensure the grid maintains stable voltages during real-time operation.

3.3 Power System Voltage Stability by Ajjarapu and Christy (1992)

They introduce the Continuation Power Flow (CPF) method to overcome convergence issues of the Newton-Raphson technique close to the voltage collapse point. As a system is loaded towards its maximum capacity, the power flow Jacobian matrix becomes increasingly ill-conditioned and eventually singular at the critical "nose point" of the P-V curve. This singularity causes conventional power flow algorithms to diverge, preventing the analysis of the system's complete voltage stability characteristics. The CPF method elegantly circumvents this problem by introducing a continuation parameter (λ), representing an incremental loading factor, and employing a predictor–corrector process to trace the full P-V curve beyond the point of Jacobian singularity. This allows for the accurate determination of the maximum loadability and the corresponding voltage stability margin.

The fundamental concept of CPF is to reformulate the power flow problem. The load and generation at buses are defined by λ, which results in a single set of equations: [6]

$$Pi = P_{Gio} - \lambda P_{Lio}$$

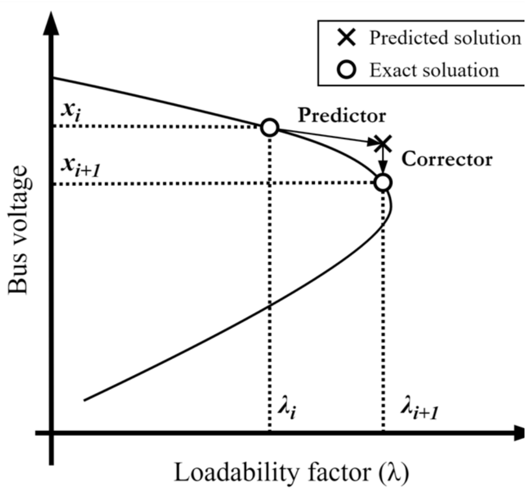
$$Qi = -\lambda P_{Lio}$$

and write the power mismatch equation using power-flow analysis as:

$$f(x, \lambda) = 0$$

where x represents the state variables (bus voltage magnitudes and angles). The solution no longer corresponds to a single point but to a continuum of points forming the P-V curve as λ varies.

Figure 4. *Structure of the typical P-V curve. Source: O. D. Atef et al., Sustainability, 2024. [7]*



The CPF constructs the P-V curve step-wise through prediction and correction. The first step is to find the tangent vector to the solution curve at the point (xi, λi). This tangent vector [dx dλ]T is obtained by solving the linear system derived from differentiating f(x,λ)=0:

$$\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial \lambda}d\lambda = 0$$

$$\begin{bmatrix} f_x & f_\lambda \end{bmatrix} \begin{bmatrix} dx \\ d\lambda \end{bmatrix} = 0$$

The predictor step then uses this tangent vector to estimate the next solution by moving a specified step size σ along this direction:

$$\begin{bmatrix} \hat{x}_{i+1} \\ \hat{\lambda}_{i+1} \end{bmatrix} = \begin{bmatrix} x_i \\ \lambda_i \end{bmatrix} + \sigma \begin{bmatrix} \frac{dx_i}{d\lambda_i} \\ 1 \end{bmatrix}$$

This provides an initial guess (x^i+1,λ^i+1). The corrector step subsequently refines this prediction by applying the Newton-Raphson method to solve the original system f(x,λ)=0, starting from this predicted point and ensuring the final solution lies precisely on the P-V curve.

3.4 Voltage Stability Evaluation Using Modal Analysis by Gao, Morison, and Kundur (1992).

Gao et al. (1992) introduce modal analysis, which uses the system's eigenvalues to evaluate voltage stability. Modal (eigenvalue) analysis linearizes the power-flow equations to identify critical modes associated with voltage instability. The smallest eigenvalue (λ_min) indicates proximity to collapse.

Starting from the Q–V sensitivity matrix:

$$\Delta Q = J_{QV}\,\Delta V$$

Eigen decomposition:

$$J_{QV} = \Phi \Lambda \Phi^{-1}$$

Smallest eigenvalue:

$$\lambda_{min} \rightarrow 0 \Rightarrow voltage\ instability$$

Bus participation factor for mode i:

$$p_{ki} = \phi_{ki}\psi_{ik}$$

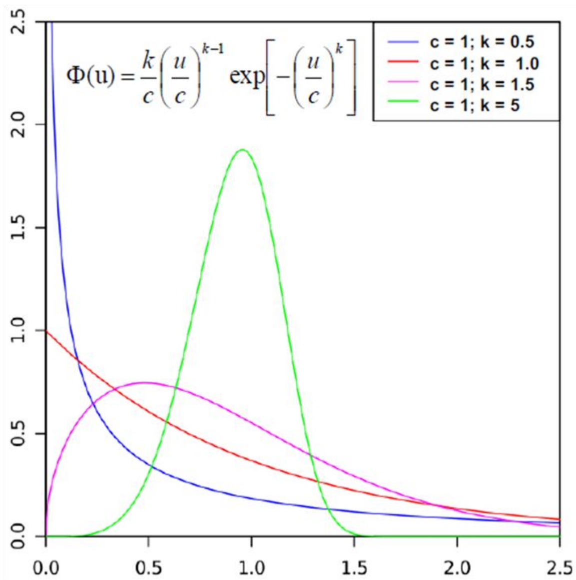
This reveals the most sensitive buses [8].

3.5 Probabilistic stability analysis: The way forward for stability analysis of sustainable power systems by Wang et al. (2017)

Wang et al. (2017) introduce probabilistic voltage-stability assessment (PVSA), which incorporates loads and uncertainties from renewable generation. Classical methods discussed above treat load and generation as deterministic, whereas PVSA incorporates uncertainty from renewable variability and stochastic demand using Monte Carlo simulations or stochastic power flow. The output is a probability distribution of voltage margins rather than a single point estimate, offering a more realistic assessment for planning under uncertainty [9].

An example of PVSA workflow can be shown combining Weibull wind-speed modeling, Monte Carlo sampling, and Continuation Power Flow (CPF) to assess how wind variability affects system voltage stability. Wind speed is represented using the Weibull probability density function below.

Figure 5. *Weibull probability density function*
Source: Calimo and P. Leitch, Wikipedia, 2008. [10]



The Weibull probability density function is used to model wind speed:

$$f_w(v) = \left(\frac{k}{c}\right)\left(\frac{v}{c}\right)^{k-1} \exp\left[-\left(\frac{v}{c}\right)^{k}\right], \qquad v \geq 0$$

Here, v is the wind speed, k is the shape parameter that controls how peaked the distribution is, and **c** is the scale parameter. We estimate the value of k and c from historical wind-speed data from the wind farm.

Using these models, Monte Carlo trials draw random wind speeds from the Weibull distribution, convert them to wind power using the turbine power curve, and combine the resulting generation with forecasted loads. Each trial then runs a CPF to obtain the P–V curve and its nose point, Pmax representing maximum system loadability. The load margin for each trial is computed as:

$$LM_n = P_{\max(n)} - P_0,$$

where, Po is the current operating load. Repeating this process across many Monte Carlo samples yields a distribution of load margins. From this distribution, system operators can estimate the expected security margin, the variability of system stability under uncertain wind, and the probability that the system load margin falls below acceptable limits. This makes the PVSA workflow a practical probabilistic tool for analyzing voltage stability under renewable uncertainty.

3.6 Synchrophasor Based Voltage Stability Assessment Using PMUs by Glavic et al. (2008), Glavic et al. (2012), and Jamei et al. (2018)

The introduction of Phasor Measurement Units (PMUs) enables real-time voltage-stability monitoring through online estimation of Thevenin equivalents or sensitivity-based indicators. Glavic and Van Cutsem et al. (2008) demonstrated that PMU measurements can detect the onset of voltage instability during large disturbances, establishing synchrophasors as a practical tool for dynamic stability assessment [11]. Subsequent work by Glavic et al. (2012) refined these ideas by introducing real-time computation and visualization of voltage-stability margins using PMU-derived Thevenin parameters, making online monitoring more intuitive for operators [12]. More recently, Jamei et al. (2018) developed techniques for continuous online tracking of Thevenin parameters from streaming PMU data, significantly improving the accuracy and responsiveness of real-time voltage-stability assessment [13]. As a result, PMU-based visualization and decision-support tools are increasingly integrated into modern energy management systems.

3.7 Machine Learning-Based Voltage-Stability Assessment by Amroune (2021), Babaali & Ameli (2023), Li et al. (2023), Guddanti et al. (2023), and Singh et al. (2023)

Machine-learning (ML) approaches have become increasingly significant for real-time or near real-time voltage-stability assessment as power systems evolve toward higher variability and widespread PMU deployment. The synthesis provided by M. Amroune (2021) shows that earlier ML techniques—such as artificial neural networks (ANNs), support vector machines (SVMs), decision trees (DTs), and adaptive neuro-fuzzy inference systems (ANFIS)—were historically used to predict voltage-stability indicators, voltage-collapse margins, and stable/unstable post-disturbance trajectories. ANNs dominated the pre-2020 literature due to their simplicity, ability to approximate nonlinear P–V relationships, and relatively low data requirements.

A particularly important subclass of ML approaches is hybrid fuzzy–neural models, of which ANFIS is the most widely used example. ANFIS combines neural-network learning with a fuzzy-logic rule base, embedding expert knowledge via membership functions while tuning rule parameters through data-driven optimization. This hybrid nature makes ANFIS a bridge between purely data-driven and physically interpretable modeling frameworks. Across these models, typical input features include bus voltage magnitudes and angles, reactive-power injections, line power flows, generator outputs, and Thévenin-based sensitivity indicators [14].

Recent literature demonstrates a shift from classical ANN-based classifiers to deep learning, ensemble learning, and topology-aware hybrid architectures. For example, A. H. Babaali and M. T. Ameli (2023) developed a weighted ensemble-learning framework for real-time short-term voltage-stability assessment (STVS) using PMU measurements. Their method integrates multiple classifiers and leverages time-series PMU inputs such as V, δ, and dynamic derivatives like dV/dt, improving robustness under contingency-driven variability [15].

Similarly, a deep transfer-learning approach by Y. Li et al. (2023) employs post-disturbance PMU trajectories—including voltage magnitude, angle, and active/reactive power (V, δ, P, Q)—along with GAN-based data augmentation to address limited labeled data. Their framework adapts effectively to changes in system topology, making it suitable for wide-area situational awareness [16].

Topology-aware deep learning has also emerged as a promising direction. The graph neural-network (GNN) model of Kishan Prudhvi Guddanti et al. (2023) embeds the electrical network directly into a graph structure, mapping bus-level PMU measurements onto graph nodes and using line connectivity as edges. By leveraging graph message-passing operations, their model predicts voltage-stability margins while maintaining generalization across switching actions and renewable-driven variability [17].

Parallel to these data-driven advancements, physics-informed machine-learning (PIML) methods have gained traction as hybrid approaches combining neural networks with physical power-system constraints. A notable example is the physics-informed transfer-learning framework by M. K. Singh et al. (2023), which embeds the steady-state power-flow constraint

$$F(x, \lambda) = 0,$$

directly into the ML training process. Using input features such as bus voltages, angles, injections, and line flows, the method ensures physically consistent predictions while enabling transfer learning across differing grid topologies or dispatch scenarios. These hybrid physics-guided models alleviate the generalization limitations observed in conventional purely data-driven methods and reduce training-data requirements [18].

Overall, the landscape of ML-based voltage-stability assessment is transitioning from classical ANN classifiers toward deep learning, ensemble methods, graph-based architectures, and physics-informed hybrid models. These approaches, leveraging PMU measurements, topology information, or embedded physical constraints, provide faster, more robust, and more interpretable assessments aligned with the needs of renewable-rich and dynamically evolving power-system environments.

4. Critical Discussion

Classical voltage-stability assessment methods such as P–V, Q–V, CPF, and modal analysis remain essential due to their strong analytical foundation and industry adoption [1], [2], [6], [7]. These techniques are transparent and effective for identifying weak buses and estimating stability margins. However, they rely on deterministic assumptions and are mainly suited for offline studies. With increasing renewable penetration and fast-changing operating conditions, their limitations become more pronounced [4].

Probabilistic methods such as PVSA address uncertainty by generating distributions of voltage margins, making them more realistic for modern systems. Despite this advantage, their high computational cost limits their feasibility for real-time applications [8].

Synchrophasor-based approaches using PMUs introduce real-time observability of voltage stability. PMU-derived Thévenin equivalents have been shown to detect instability onset and support operator decision-making [9]–[11]. Their performance, however, depends on PMU placement, communication latency, and measurement quality.

Machine-learning methods have gained significant traction due to their ability to capture nonlinear relationships and provide fast stability predictions using wide-area measurements [14]. Earlier ML approaches such as ANN, SVM, DT, and ANFIS demonstrated reasonable performance but often required extensive feature engineering and struggled to generalize across varying conditions. Recent advancements—including weighted ensemble learning, deep transfer learning, graph-based neural networks, and physics-informed hybrid models—address many of these limitations by leveraging PMU time-series, topology information, or embedded power-system constraints [15]–[18]. These methods provide improved robustness to renewable variability, adaptability under topology changes, and enhanced physical consistency. Nevertheless, they still rely on representative training datasets, stable PMU availability, and periodic model updates as system conditions evolve.

Overall, the field is moving toward real-time, uncertainty-aware, and hybrid frameworks that integrate classical analysis, probabilistic modeling, PMU-based measurement, and ML-driven prediction.

To summarize these findings, Table I presents a comparative overview of all reviewed voltage-stability assessment methods, including their principles, outputs, advantages, and limitations.

Table I. *Comparative evaluation of voltage-stability assessment paradigms and industry practices.* [1-2, 5-6, 8-9, 11-18]

Approach	Nature	Primary Advantage	Key Limitation	Industry Practices
P–V & Q–V Curves	Deterministic	Simple visualization	Not suitable for real-time	Standard in utility planning
CPF	Numerical continuation	Accurate & robust	Computational cost	Widely used in planning software
Modal Analysis	Analytical	Deep system insight	Complex implementation	Planning & research
PVSA	Probabilistic	Handles uncertainty	Time-consuming	Pilot-level research
PMU-Based	Real-time	Online monitoring	Data-intensive	Increasing adoption
AI/ML	Data-driven/ Mixed	Fast, predictive	Requires datasets	Emerging pilot projects

5. Future Directions and Applications

Future research in voltage-stability assessment will focus on integrating deterministic and data-driven paradigms into unified, real-time frameworks. Combining continuation power flow with synchrophasor-based data analytics will allow operators to monitor and predict voltage margins dynamically. AI-enhanced reactive-power controllers using reinforcement learning will enable adaptive system responses to voltage fluctuations. Probabilistic approaches will quantify stability risk under renewable uncertainty, improving grid planning and resilience. Finally, standardized interoperability between IEEE, NERC, and IEC frameworks will ensure consistency in modeling and validation of new methods across global power systems.

6. Conclusion

Voltage stability remains one of the most critical aspects of secure grid operation. Classical deterministic methods such as P–V, Q–V, CPF, and modal analysis have laid a strong theoretical foundation but are increasingly complemented by probabilistic, PMU-based, and AI-driven approaches. As power systems transition toward renewable and inverter-based architectures, hybrid and data-driven frameworks promise real-time and uncertainty-aware assessment capabilities. These innovations will enhance the predictive and preventive capacity of future power networks, ensuring stable and resilient operation under diverse conditions.

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